

Exam Date & Time: 06-Jan-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - JANUARY-2024

Principles of Computer Vision [CSE-3171]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) With a necessary graph, explain the Power-law and Image Log transformations used in image enhancement. What is the necessity of such image enhancements? (4)

A)

- B) Filter the image shown below by applying 3x3 median filter mask and justify that median filtering minimizes salt and pepper noise. Assume zero padding in the border.

24	23	33	25	32	24
34	255	24	0	26	23
23	21	32	31	28	26

(3)

- C) A matrix representation of the image obtained by surveillance camera is shown below. Answer the following with respect to the given image:

- Show the intensity distribution of this image using histogram
- Compute the contrast of the image
- Identify whether the obtained image is of high or low contrast.

200	200	200	180	240
180	180	180	180	190
190	190	190	190	180
190	200	220	220	240
230	180	190	210	230

(3)

- 2) How Laplacian technique can be used to detect horizontal and vertical edges in the image? Explain with appropriate masks and mathematical equations. (5)

A)

- B) Consider the image matrix given below. Select the threshold and perform Global threshold-based segmentation on the above image. Show the output matrix. (5)

128	128	128	64	64	32	32	8
64	64	128	128	128	8	32	32
32	8	64	128	128	64	64	64
8	128	128	64	64	8	64	64
128	64	64	64	128	128	8	8
64	64	64	128	128	128	32	32
8	128	32	64	64	128	128	128
8	8	64	64	128	128	64	64

- 3) Represent the following camera models with a neat diagram and describe how the scene image is projected onto a view plane.
i) Thin lens camera model ii) Pinhole camera model (5)
- A)
- B) What are the different applications of stereo correspondence? Suppose there are two cameras which are focusing on the same scene. Apply Epipolar geometry to identify the distance between two camera centers and the orientation of image planes with respect to each other. (5)
- 4) What is the goal of object tracking in computer vision? Explain how to handle the low contrast and noisy video frames in object tracking. (3)
- A)
- B) Illustrate how four different error metrics can be applied in estimating the motion using translational alignment. What is the drawback of each error metric? (4)
- C) Differentiate between layered motion estimation, Parametric motion estimation and translational motion estimation techniques. (3)
- 5) Explain different phases of computer vision involved in face recognition in the crowd. (5)
- A)
- B) A content-based image retrieval system retrieves 10 similar images from the database based on color of the object in the image. Explain the different steps of image processing that can be in this system. (5)

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